

## Problem

The natural response of an  $RLC$  circuit is described by the differential equation

$$\frac{d^2v}{dt^2} + 2\frac{dv}{dt} + v = 0$$

for which the initial conditions are  $v(0) = 350$  V and  $dv(0)/dt = 0$ . Solve for  $v(t)$ .

## Step-by-step solution

## Step 1 of 3

The second order differential equation of  $RLC$  circuit is,

$$\frac{d^2v}{dt^2} + 2\frac{dv}{dt} + v = 0 \quad \dots\dots (1)$$

The initial conditions are,  $v(0) = 350$  V and  $\frac{dv(0)}{dt} = 0$ .

Applying Laplace Transform to the equation 1,

$$\left[ s^2V(s) - sv(0) - \frac{dv(0)}{dt} \right] + 2[sV(s) - v(0)] + V(s) = 0$$

Substitute 350 V for  $v(0)$  and 0 for  $\frac{dv(0)}{dt}$ .

$$[s^2V(s) - 350s - 0] + 2[sV(s) - 350] + V(s) = 0$$

$$[s^2 + 2s + 1]V(s) - 350s - 700 = 0$$

$$\begin{aligned} V(s) &= \frac{350s + 700}{s^2 + 2s + 1} \\ &= \frac{350(s + 2)}{(s + 1)^2} \end{aligned}$$

## Step 2 of 3

Take partial fractions.

$$V(s) = \frac{A}{s+1} + \frac{B}{(s+1)^2} \quad \dots\dots (2)$$

$$\frac{350(s+2)}{(s+1)^2} = \frac{A}{s+1} + \frac{B}{(s+1)^2}$$

$$350s + 700 = A(s+1) + B$$

By comparing coefficients of  $s$ ,

$$A = 350$$

By comparing coefficients of constants,

$$A + B = 700$$

$$A + 350 = 700$$

$$A = 350$$

## Step 3 of 3

Substitute 350 for  $A$  and 350 for  $B$  in equation 2.

$$V(s) = \frac{350}{s+1} + \frac{350}{(s+1)^2}$$

Apply Inverse Laplace Transform.

$$L^{-1}[V(s)] = L^{-1}\left[\frac{350}{s+1}\right] + L^{-1}\left[\frac{350}{(s+1)^2}\right]$$

$$\begin{aligned}v(t) &= 350e^{-t}u(t) + 350te^{-t}u(t) \\&= 350(1+t)e^{-t}u(t) \text{ V} \\&= 350(1+t)e^{-t} \text{ V}, \quad t > 0\end{aligned}$$

Thus, the  $v(t)$  for  $t > 0$  is  $\boxed{350(1+t)e^{-t} \text{ V}}$ .